

ENTERPRISE BUSINESS INTELLIGENCE

Root Cause and Profitability Analysis

Executive Management Report

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Role	Data Analyst
Analysis Period	September 2025 to August 2026
Datasets Analyzed	18 enterprise datasets
Tools Used	MySQL, Python, Power BI and Excel
Report Generated	16 September 2026

Confidential - Prepared for Senior Management

1. Executive Summary

The analysis confirms that the company has a strong revenue base, but it is not converting revenue into profit. Total analyzed revenue was INR 48.43 crore, while the company recorded a net loss of INR 3.25 crore and a net profit margin of -6.70%.

KPI	Result	Management Interpretation
Revenue	INR 48.43 Cr	Strong sales volume
Net Profit	INR -3.25 Cr	Company is loss-making
Profit Margin	-6.70%	Cost structure is unsustainable
Revenue Reconciliation Gap	INR 3.90 Cr	Revenue reports do not reconcile
Overdue Receivables >180 Days	INR 4.52 Cr	High collection and credit risk
Excess Inventory Value	INR 79.77 Cr	Significant working-capital blockage
Supplier On-Time Delivery	11.22%	Supplier performance is unacceptable
Next-Quarter Revenue Forecast	INR 12.31 Cr	98.15% of target; target is at risk

Primary conclusion: Revenue growth alone is not creating business value. Product cost, marketing cost, excess inventory, supplier delays, returns, discounts and collection delays are reducing profitability and cash flow.

2. Data Quality Investigation

An automated data-quality audit was performed across all 18 datasets. The audit checked duplicate records, missing keys, invalid dates, future dates, negative quantities, invalid prices, unmatched records and broken foreign-key relationships.

Dataset	Records
Customer Complaints	10,036
Customer Master	19,354
Employee Attendance	49,444
Employee Master	499
Inventory	118,469
Invoices	101,080
Marketing Campaigns	4,941
Monthly Targets	240
Order Items	245,340
Orders	99,266
Payments	76,362
Product Master	992
Purchases	23,588
Region Master	20
Returns	14,751
Sales	244,188
Suppliers	200
Website Activity	198,400

The audit identified duplicate records, missing values, invalid zero dates, unmatched references and negative inventory. Cleaning and quality scoring were completed using automated scripts without manually modifying individual source records.

3. Revenue Reconciliation

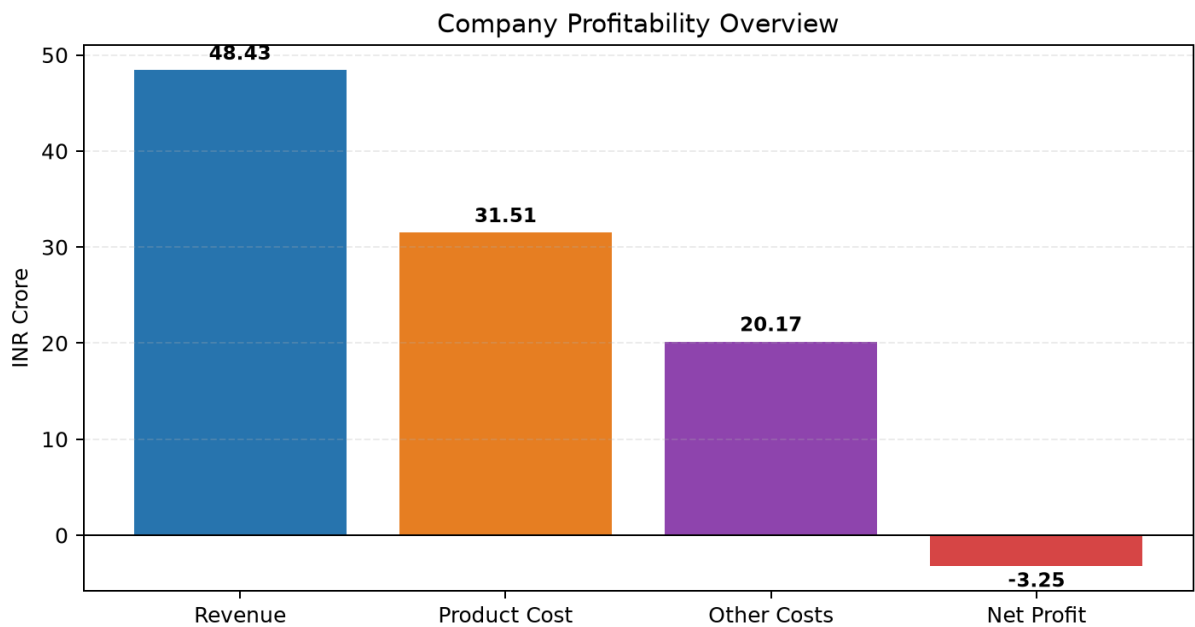
Reconciliation Component	Result
Management Revenue	INR 52.00 Cr
Database Gross Sales	INR 49.92 Cr
Sales Revenue after adjustment	INR 48.43 Cr
Returns	INR 1.50 Cr
Refunds	INR 0.32 Cr
Unmatched Transactions	798 records
Final Reconciliation Difference	INR 3.90 Cr

Root cause: Management and database reports use different revenue definitions. Unmatched transactions and inconsistent treatment of returns, refunds, invoices and payments contribute to the difference.

Recommended action: Establish one governed revenue definition and automate order-to-cash reconciliation before publishing monthly management reports.

4. Profitability Investigation

Profitability Component	Amount
Revenue	INR 48.43 Cr
Discounts	INR 1.50 Cr
Refunds	INR 0.32 Cr
Product Cost	INR 31.51 Cr
Shipping Cost	INR 1.59 Cr
Payment Fees	INR 0.87 Cr
Allocated Marketing Cost	INR 17.38 Cr
Net Profit	INR -3.25 Cr
Net Profit Margin	-6.70%



Marketing cost and product cost are the largest profitability pressures. All major categories and regions generated negative net profit. Approximately 990 products were classified as negative-profit products.

5. Customer and Cohort Analytics

Customer Segment	Customers	Revenue	Net Profit	Margin
At Risk	7,834	INR 18.73 Cr	INR -1.27 Cr	-6.79%
Regular	6,963	INR 14.94 Cr	INR -1.09 Cr	-7.27%
Low Value	3,905	INR 11.79 Cr	INR -0.70 Cr	-5.96%
High Value	1,007	INR 2.89 Cr	INR -0.18 Cr	-6.16%
Churned	46	INR 0.07 Cr	INR -0.005 Cr	-7.75%

Cohort Retention Findings

Cohort	Customers	Month-1 Retention	Trend
September 2025	12,152	42.59%	First cohort
October 2025	7,245	40.29%	Declining
November 2025	340	1.18%	Sharp decline
December 2025	14	50.00%	Small cohort
January 2026	4	25.00%	Small cohort

The November 2025 cohort experienced the largest retention decline. December 2025 and January 2026 contain very small cohorts and should not be used for strong management conclusions.

6. Product and Marketing Analysis

Product Finding	Result
Products Analyzed	992
Negative-Profit Products	990
Highest-Revenue Category	Accessories - INR 8.27 Cr
Category Profitability	All major categories negative
Category Margin Range	Approximately -6.65% to -6.75%
Main Product Issue	Revenue increasing without positive profit

Marketing Attribution

Channel	Linear Revenue	Linear Profit	Profit ROI
Instagram	INR 82.51 L	INR -5.50 L	-1.60%
Google Ads	INR 79.69 L	INR -5.51 L	-1.63%
Email	INR 85.80 L	INR -5.70 L	-1.64%
Organic Search	INR 83.89 L	INR -5.73 L	-1.60%
Referral	INR 88.59 L	INR -5.78 L	-1.64%

Instagram produced the least negative attributed profit. However, no marketing channel produced positive attributed profit. Therefore, Instagram should not be described as a profitable channel.

7. A/B Test Investigation

Metric	Control	Treatment
Sessions	50,000	49,993
Conversions	4,302	4,328
Conversion Rate	8.6040%	8.6572%
95% Group Confidence Interval	8.3582% to 8.8498%	8.4107% to 8.9037%

Statistical Measure	Result
Absolute Lift	0.0532 percentage points
Relative Lift	0.6185%
Z Statistic	0.299601
P-Value	0.764481
95% CI for Difference	-0.2949% to 0.4013%
Cohen's H	0.001895 - negligible effect
Statistical Decision	Not statistically significant

Management decision: Do not permanently implement the treatment checkout design yet. Continue testing because the observed improvement is extremely small and the p-value is greater than 0.05.

8. Inventory Investigation

Inventory Metric	Result
Inventory Turnover	0.12 times
Days Inventory	2,966.72 days
Stockout Rate	11.66%
Critical Products	9
Dead-Stock Products	62
Negative Inventory Products	120
Excess Inventory Units	123,207.55
Excess Inventory Value	INR 79.77 Cr

The company simultaneously holds substantial excess inventory and experiences stockout risk for high-demand products. This indicates ineffective demand planning and reorder policies.

Recommended action: Freeze purchases for dead-stock products, liquidate excess stock, correct negative inventory records and introduce demand-based reorder points using supplier lead time.

9. Supplier Analytics

Supplier Metric	Result
Total Suppliers	200
Total Purchase Value	INR 4,066.00 Cr
Average Delivery Time	24.46 days
Overall On-Time Delivery	11.22%
Overall Rejection Rate	0.81%
Average Supplier Score	54.97
Suppliers Requiring Review	121
Best Supplier	Supplier 075 - Score 68.34
Lowest-Ranked Supplier	Supplier 134 - Score 38.89
Declining Supplier	Supplier 061

Supplier 061 demonstrated increasing cost with declining delivery and quality indicators. Supplier 134 received the lowest overall performance score and requires formal review.

Recommended action: Renegotiate pricing and service-level agreements, introduce delivery penalties, perform quality audits and shift critical purchase volume to better-performing suppliers.

10. Forecasting and Target Achievement

Forecast Month	Revenue Forecast	Revenue Target	Achievement	Status
September 2026	INR 4.10 Cr	INR 4.36 Cr	93.89%	At Risk
October 2026	INR 4.10 Cr	INR 4.18 Cr	98.15%	At Risk
November 2026	INR 4.11 Cr	INR 4.00 Cr	102.80%	Above Target
Next Quarter	INR 12.31 Cr	INR 12.55 Cr	98.15%	At Risk

The next-quarter revenue forecast is INR 12.31 crore against a target of INR 12.55 crore. Forecast achievement is 98.15%, leaving a forecast shortfall of approximately INR 0.23 crore.

The forecast uses only 12 months of monthly history. Seasonality, promotions, market changes, pricing decisions and data corrections may materially affect actual performance.

11. Multi-Factor Anomaly Detection

Dataset	Total Records	Model Flagged	Critical / High Risk	Maximum Score
Sales	246,370	2,464	1,456 critical	97.24
Payments	76,510	766	9,764 high-risk	89.36
Inventory	119,040	1,188	3,807 rule-critical	99.09

The anomaly score used transaction amount, transaction frequency, historical customer behaviour, product behaviour, discount and refund behaviour, time patterns, inventory movements and an Isolation Forest machine-learning score.

The highest-risk records involved negative closing stock, unusual inventory movement, abnormal stock-value changes and negative inventory value. The maximum observed anomaly score was 99.09.

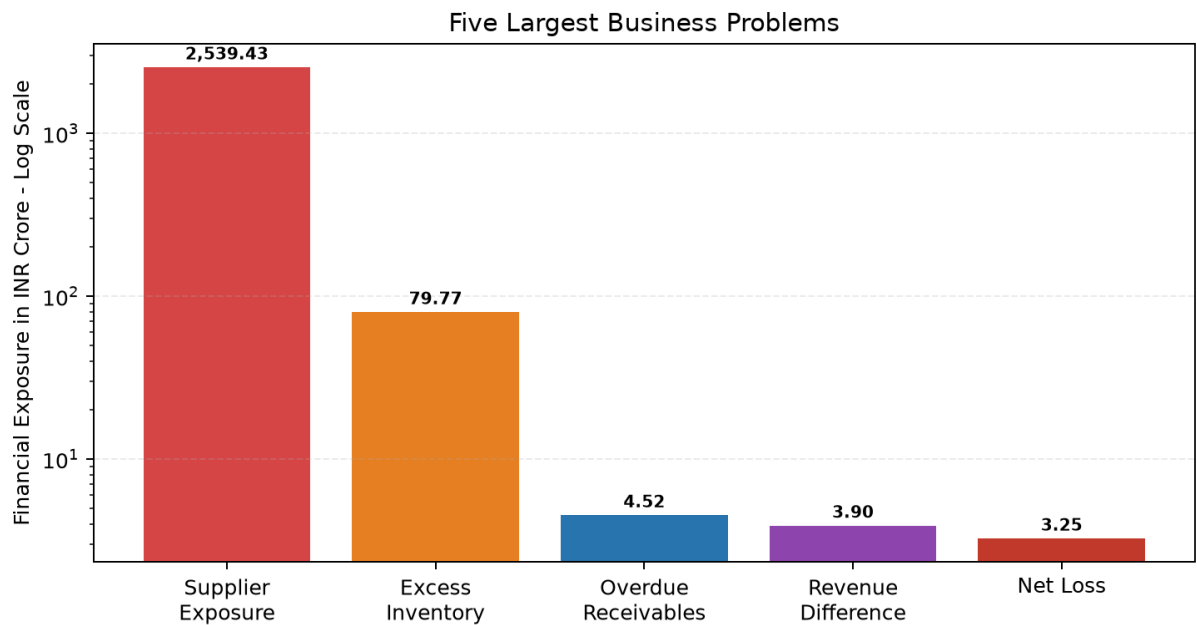
12. Executive Exception Engine

Exception Metric	Result
Total Exceptions	14,367
Critical Exceptions	2,122
High-Priority Exceptions	12,245
Negative-Profit Products	990
Critical Inventory Products	9
Negative Inventory Products	120
Unique Overdue Invoices >180 Days	11,771
Total Overdue Amount >180 Days	INR 4.52 Cr
Suppliers Requiring Review	121
Critical Anomalies	1,000

Gross exception exposure contains overlapping business issues and must not be interpreted as unique company loss. The same inventory, supplier or product exposure may appear under multiple exception types.

13. Five Largest Business Problems

Rank	Business Problem	Actual Result	Impact / Exposure
1	Revenue generated but company remains unprofitable	INR 3.25 Cr loss	INR 3.25 Cr
2	Excess inventory and extremely low turnover	INR 79.77 Cr excess	INR 79.77 Cr
3	Supplier delivery performance is unacceptable	11.22% on-time	INR 2,539.43 Cr purchase exposure
4	Revenue reports do not reconcile	INR 3.90 Cr gap	INR 3.90 Cr
5	Long-overdue customer receivables	INR 4.52 Cr overdue	INR 4.52 Cr



Supplier purchase exposure represents the purchase value influenced by supplier-performance issues. It is not equivalent to realized financial loss.

14. Management Recommendations

Priority	Recommended Action	Expected Benefit
Critical	Correct loss-making prices and control product, shipping, payment and marketing costs.	Restore positive contribution margin.
Critical	Automate order-to-cash reconciliation using one governed revenue definition.	Eliminate inconsistent revenue reporting.
Critical	Freeze dead-stock purchases and introduce demand-based reorder points.	Release working capital and reduce stock risk.
Critical	Renegotiate low-performing suppliers and enforce service-level penalties.	Improve availability and delivery reliability.
High	Create ageing-based collection workflows and review customer credit limits.	Reduce overdue receivables and improve cash flow.
High	Redesign marketing campaigns that generate negative attributed profit.	Improve customer acquisition economics.
High	Investigate critical anomalies before transaction approval or reporting.	Reduce fraud, errors and operational leakage.
Medium	Continue the checkout A/B test using a predefined sample-size requirement.	Avoid unsupported implementation decisions.

15. Assumptions, Limitations and Conclusion

- 1. The source data intentionally contained duplicates, missing values, invalid dates and reconciliation problems.
- 2. Financial results depend on the available fields and the analytical calculation logic used in this project.
- 3. Allocated marketing cost is an analytical allocation and may differ from accounting treatment.
- 4. Forecasting uses only 12 months of monthly history and should be refreshed as more data becomes available.
- 5. Supplier purchase exposure and gross exception exposure are not equivalent to realized financial loss.
- 6. A/B test conclusions apply only to the available experiment population and test period.
- 7. Cohorts containing very few customers should not drive major management decisions.

Final conclusion: The company has a strong revenue base, but cost control, inventory planning, supplier performance, collections and data governance require immediate improvement. Management should prioritize profitability and cash conversion instead of focusing only on revenue growth.

Supporting evidence is available in the Data Quality, SQL, Python, Power BI, Reconciliation, Exception, Forecasting, Inventory, Marketing, Supplier, Anomaly and Root Cause Analysis folders.